MAT1575, Classwork10, Fall2026

3.4 Partial Fractions (p.262-270)

1. Evaluate each of the following integrals:

(a) $\int \left(\frac{1}{x+1}+\frac{2}{x-2}\right)dx$

$= \int \frac{dx}{x+1} + \int \frac{2dx}{x-2}$

$= \ln|x+1| + 2\ln|x-2| + C$

(b) $\int \frac{3x}{x^2-x-2}dx = \int \frac{3x}{(x-2)(x+1)}dx$

$3x^2+2x+1$ deg 2, leading coeff 3

$5x+4x^3+2x+5$ deg 3 4

2. Rational Function. A rational function is a fraction $\frac{f(x)}{g(x)}$ where $f(x)$ and $g(x)$ are both polynomials, and $g(x) \neq 0$ is not the zero function.

3. Degree of a Polynomial. The coefficient of the variable to the highest power in a polynomial $f(x)$, is called the leading coefficient and the highest power is called the degree of the polynomial, denoted by $\deg(f)$.

4. Example when $\deg(f(x)) \geq \deg(g(x))$: Evaluate $\int \frac{x^2+3x+5}{x+1}dx$ ($\deg 2$ / $\deg 1$)

$\int \frac{x^2+3x+5}{x+1}dx = \int \left(x+2+\frac{3}{x+1}\right)dx$

$= \frac{x^2}{2} + 2x + 3\ln|x+1| + C$

$x+2$

$x+1 \;\big)\; x^2+3x+5$

$-(x^2+x)$

$2x+5$

$-(2x+2)$

$\frac{x^2+3x+5}{(x+1)} = \frac{(x+1)(x+2)+3}{(x+1)} = \frac{(x+1)(x+2)}{(x+1)} + \frac{3}{(x+1)}$

5. Partial Fraction Decomposition: To evaluate $\int \frac{f(x)}{g(x)}dx$ with $\deg(f) < \deg(g)$, we can separtate the fraction $\frac{f(x)}{g(x)}$ into several rational functions and then solve them one by one.

6. Partial Fraction with Nonrepeated Linear Factors: $\frac{f(x)}{g(x)} = \frac{A_1}{a_1x+b_1} + \frac{A_2}{a_2x+b_2} + \cdots + \frac{A_n}{a_nx+b_n}$

Evaluate $\int \frac{3x+2}{x^3-x^2-2x}dx$.

$= \int \frac{3x+2}{x(x-2)(x+1)}dx$

$= \int \frac{-1}{x} + \frac{\frac{4}{3}}{x-2} + \frac{-\frac{1}{3}}{x+1}dx$

$= -\ln|x| + \frac{4}{3}\ln|x-2| - \frac{1}{3}\ln|x+1| + C$

$x^3-x^2-2x = x(x^2-x-2) = x(x-2)(x+1)$

$\frac{3x+2}{x(x-2)(x+1)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+1}$

$= \frac{A(x-2)(x+1) + Bx(x+1) + Cx(x-2)}{x(x-2)(x+1)}$

$3x+2 = A(x-2)(x+1) + Bx(x+1) + Cx(x-2)$

To find A, try $x=0$, $2 = A(0-2)(0+1) \Rightarrow 2 = -2A$, $A = -1$

To find B, try $x=2$, $3\cdot 2+2 = 0 + B\cdot 2\cdot(2+1) \Rightarrow B = \frac{4}{3}$

To find C, try $x=-1$, $3(-1)+2 = 0 + 0 + C\cdot(-1)(-1-2)$, $C = \frac{-1}{3}$

7. Partial Fraction with Repeated Linear Factors: $\frac{f(x)}{g(x)} = \frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_n}{(ax+b)^n}$

Evaluate $\int \frac{x-2}{(2x-1)^2(x-1)}\,dx$.

$\frac{x-2}{(2x-1)^2(x-1)} = \frac{A}{(x-1)} + \frac{B}{(2x-1)} + \frac{D}{(2x-1)^2}$

$= \frac{A(2x-1)^2 + B(x-1)(2x-1)}{(x-1)(2x-1)^2}$

$= \int \frac{-1}{(x-1)} + \frac{2}{(2x-1)} + \frac{3}{(2x-1)^2}\,dx$

$= -\ln|x-1| + \frac{2\ln|2x-1|}{2} - \frac{3}{2}\frac{1}{2x-1} + C$

$x-2 = A(2x-1)^2 + B(x-1)(2x-1) + D(x-1)$

To find A, $x=1$, $1-2 = A\cdot(2\cdot 1-1)^2 \Rightarrow A=-1$

To find D, $x=\frac{1}{2}$, $\frac{1}{2}-2 = 0+0+D\cdot(\frac{1}{2}-1)$

$-\frac{3}{2} = -\frac{1}{2}D \Rightarrow D=3$

To find B, try $x=0$, $0-2 = (-1)\cdot(2\cdot 0-1)^2 + B(0-1)(2\cdot 0-1) + 3\cdot(-1)$

$-2 = -1 + B - 3$, $B=2$

$du = 2x-1$, $u = 2dx$, $\frac{u\,du}{2} = dx$

$\int \frac{2\,dx}{(2x-1)} = \int \frac{2}{u}\frac{du}{2} = \ln|u| + C$

8. Partial Fraction with Irreducible Quadratic Factors: $\frac{f(x)}{g(x)} = \frac{A_1x+B_1}{a_1x^2+b_1x+c_1} + \frac{A_2x+B_2}{a_2x^2+b_2x+c_2} + \cdots + \frac{A_nx+B_n}{a_nx^2+b_nx+c_n}$

Evaluate $\int \frac{2x-3}{x^3+x}\,dx$.

9. Express the rational function as a sum or difference of two or several simpler rational expressions.

(a) $\frac{x-3}{x+2}$ (b) $\frac{x^2+3x+1}{x^2-4}$ (c) $\frac{x+1}{(x+3)(x-2)}$ (d) $\frac{x-2}{(x+1)(x^2+4)}$ (e) $\frac{2x+3}{(x^2+1)(x^2+4)}$